

Hierarchical CMR for Across-Bridge Equivalence

Changes from v5

- **Dropped Version B.** The Stage 2 empirical check (re-scoring pilot Exp 2 responses against the within-storyline-bridge target) confirmed Version A's sharp prediction: Exp 2 participants produce a decent number of within-storyline-neighbor responses. The Exp 2 « Exp 3 gap is a scoring effect, not a retrieval-routing effect. The μ parameter is no longer needed and has been removed.
- **Collapsed retrieval to a single section.** The former §3 (Version A retrieval) is now the only retrieval section. The former §4 (Version B retrieval) is gone.
- **Pruned the parameter inventory to what's actually doing work in this paradigm.** Removed β_{rec} and ϵ_d — both are standard CMR parameters inherited without assessment, but they have no causal role in single-response cued recall (β_{rec} governs context drift between consecutive recalls; ϵ_d governs a stopping rule that doesn't apply when there's exactly one response per trial). Added §5.1 explaining their absence and flagging that they would be reintroduced if the paradigm is extended to include free recall. Total: 4 inherited CMR parameters + 1 new (λ).
- **Reframed §6.2** from "decisive test between versions" to "empirical validation of the scoring-filter account" (completed).
- **Tightened the Purpose section:** the model makes exactly one substantive addition to Cornell & Zhang. No version fork.
- **Updated §8 (differences from C&Z)** to a single-column structure.
- **Updated §9 (risks):** the Version A falsification risk is no longer live. Semantic-overlap absorption and λ identifiability remain the main concerns.

Changes from v4

- Added §1.5.1 on prior work for pre-experimental context representations, situating the embedding-based $M_{\text{pre}}^{\text{FC}}$ choice within an existing lineage (Polyn et al. 2009; Morton & Polyn 2016; Heusser et al. 2021; Xu et al. 2024).

Changes from v3

- Added §1.5 on pre-experimental context; added §7.1 on Cornell 2023 parallel; added identity-vs-embedding diagnostic to §6.1.

Purpose

Explain why cued recall performance in the **across-event-bridge** condition (Exp 3; interleaved encoding, within-storyline retrieval) equals performance in the **across-event-within-storyline** condition (Exp 1; grouped encoding), despite temporally non-adjacent targets in Exp 3. The

model makes one substantive addition to Cornell & Zhang (2025): encoding-time reinstatement of a cached storyline context whenever a storyline resumes after interruption. All other mechanisms — encoding drift, boundary synchronization, retrieval — are inherited from existing CMR-family models unchanged.

Core theoretical claim. Storyline structure is encoded into memory via context reinstatement at encoding. Retrieval is a standard context-driven readout (following Cornell et al. 2024). Experimental instructions act as a response-scoring filter, not as a modulator of retrieval dynamics. Under this account, the Exp 2 $\ll$ Exp 3 gap is a scoring artifact, not a memory-accessibility effect — a prediction validated in pilot response content (§6.2).

1. Notation and primitives

Let N denote the number of scenes (items) in the stimulus. Each scene i has a pre-experimental input vector $c_i^{\text{IN}} \in \mathbb{R}^N$ (one-hot under the standard distinctiveness assumption, or a pre-trained USE embedding for naturalistic stimuli — see §1.5).

Define two context vectors, both in $\mathbb{R}^N$, evolving over encoding time i :

- c_i^{item} : item-level context (fast drift)
- c_i^{story} : storyline-level context (slow drift)

Each scene is labeled with an event index $e(i)$ and a storyline index $s(i)$. Event boundaries are defined by $e(i) \neq e(i-1)$; storyline switches by $s(i) \neq s(i-1)$. The scalar ρ is set on each update so that the context vector has unit norm: $\rho_x = \sqrt{1 - \beta_x^2}$ for each context level x .

The model maintains two associative matrices built during encoding:

- $M^{IC} \in \mathbb{R}^{N \times N}$: item $\rightarrow$ item-context associations (as in CMR)
- $M^{SC} \in \mathbb{R}^{N \times N}$: storyline-level context $\rightarrow$ item-level context associations, formed at storyline switches and storyline returns

Within-storyline event boundaries do *not* have their own associative matrix — they are handled by the boundary synchronization mechanism inherited from Cornell & Zhang, without any storage of event-level information.

1.5 Pre-experimental context

The input c_i^{IN} that drives drift in Eqs. 1 and 2 is not raw item identity — it is the *context induced* by presenting item i , which reflects both the participant's prior knowledge about the item and any associations formed during the current experiment. Following standard CMR:

$$c_i^{\text{IN}} = (1 - \gamma_{fc})M_{\text{pre}}^{FC}f_i + \gamma_{fc}M_{\text{exp}}^{FC}f_i \quad (1.5.1)$$

where f_i is the one-hot vector indicating item i , M_{pre}^{FC} is the pre-experimental item-to-context matrix, M_{exp}^{FC} is the experimental (learned) item-to-context matrix built up Hebbian-style during

encoding, and γ_{fc} is the mixing weight.

M_{pre}^{FC} encodes what each item "means" before the experiment begins. The choice of how to populate it is a modeling decision with empirical consequences:

Option 1 — identity matrix (distinctiveness assumption). Each scene's pre-experimental context is orthogonal to every other scene's. No semantic overlap between scenes is represented. Used in Cornell & Zhang (2025) and Cornell et al. (2024) to simplify analysis of list-learning paradigms where semantic structure is not the phenomenon of interest.

Option 2 — semantic similarity matrix. M_{pre}^{FC} is populated with semantic associations between items, e.g., from LSA or word co-occurrence statistics. Used in Polyn et al. (2009) to capture semantic clustering in free recall.

Option 3 — pre-trained embeddings. For naturalistic stimuli, M_{pre}^{FC} can be built from neural embeddings of each scene's content (e.g., USE embeddings of subtitle text, consistent with the pipeline in Xu et al. 2024). Scenes with overlapping content — shared characters, settings, themes — induce overlapping pre-experimental contexts.

For naturalistic film, **Option 3 is the empirically appropriate choice.** Scenes in the same storyline share characters, settings, and ongoing plot concerns, which makes their USE embeddings substantially more similar than scenes in different storylines. This has a direct consequence: *even without any storyline-level context mechanism*, scenes from the same storyline will be encoded in nearby regions of item-context space, simply because their pre-experimental inputs are already similar.

A null hypothesis the model must rule out

This creates an important alternative account of the $\text{Exp 1} \approx \text{Exp 3}$ equivalence:

The equivalence arises entirely from pre-experimental semantic overlap between scenes in the same storyline. Storyline-level context reinstatement (λ) is not doing any work; the apparent "storyline reinstatement" effect is just CMR-style retrieval picking up on semantic similarity that was already present in the stimuli.

If this account is correct, a version of the model with $\lambda = 0$ but Option-3 pre-experimental context would already reproduce $\text{Exp 1} \approx \text{Exp 3}$, and the λ parameter would be superfluous.

The Stage 1 simulation plan (§6.1) explicitly tests this by running the model twice — once with the identity matrix and once with USE embeddings — to quantify how much explanatory work pre-experimental semantic overlap does on its own.

1.5.1 Prior work on pre-experimental context representations

The choice to populate M_{pre}^{FC} with learned semantic representations has direct precedent in the CMR literature, though the specific combination of sentence-level neural embeddings with CMR for naturalistic stimuli is not yet a mainstream published approach. Worth situating:

- **Polyn, Norman & Kahana (2009)** — the foundational CMR paper — use LSA-derived word-to-word associations to populate the pre-experimental matrix, enabling CMR to capture semantic clustering in free recall of word lists. This establishes that M_{pre}^{FC} is a substantive modeling choice with empirical consequences, not a formal placeholder.

- **Morton & Polyn (2016, *Cognitive Psychology*)** — the clearest methodological precedent — systematically compare three semantic-structure models (LSA, GloVe, word association spaces) embedded within CMR, evaluated by how well each predicts recall-order likelihoods. They find differential predictive power across representations, demonstrating that the choice of embedding matters empirically and can be tested against data.
- **Heusser, Fitzpatrick & Manning (2021, *Nature Human Behaviour*)** — the "Sherlock paper" — analyze naturalistic movie memory by embedding scenes and recall events in a shared topic-model space, then comparing trajectories through that space. This is not a CMR-family model, but establishes the methodological move of using sentence-level embedding representations to analyze naturalistic memory, and provides pipeline infrastructure the present work inherits from.
- **Xu, Zhu & Manning (2024, *Nature Communications*)** — directly relevant — use USE embeddings of scene subtitles to automate matching of free-recall responses to video scenes, achieving 98.6% classification accuracy against manual coding. The present work extends this same representation from a scoring tool to a model input, using USE embeddings as M_{pre}^{FC} .

The methodological contribution of the present model is therefore modest but real: extending Morton & Polyn's approach of embedding semantic structure within CMR from word-level LSA/GloVe representations to scene-level neural sentence embeddings, and applying this in combination with the hierarchical context machinery from Cornell & Zhang (2025). Framed this way, the embedding choice is a principled extension of established CMR practice, not a novel claim requiring separate justification.

2. Encoding dynamics

2.1 Within-event, within-storyline drift

When scene i appears and neither an event nor storyline boundary has been crossed ($e(i) = e(i-1)$, $s(i) = s(i-1)$), both contexts drift toward the input:

$$c_i^{\text{item}} = \rho_{\text{item}} c_{i-1}^{\text{item}} + \beta_{\text{enc}} c_i^{\text{IN}} \quad (1)$$

$$c_i^{\text{story}} = \rho_{\text{story}} c_{i-1}^{\text{story}} + \beta_{\text{story}} c_i^{\text{IN}} \quad (2)$$

with drift rates ordered $\beta_{\text{enc}} > \beta_{\text{story}}$. The item-level context accumulates item-level associations exactly as in CMR:

$$\Delta M^{IC} = c_i^{\text{item}} f_i^\top \quad (3)$$

where f_i is the item representation at position i .

2.2 Event boundary within a storyline

When $e(i) \neq e(i-1)$ but $s(i) = s(i-1)$, the item-level context synchronizes to the current storyline context (Cornell & Zhang's boundary synchronization):

$$c_i^{\text{item}} \leftarrow c_i^{\text{story}} \quad (4)$$

The storyline context is unaffected — it continues to drift per Eq. 2.

2.3 Storyline switch

When $s(i) \neq s(i-1)$, the outgoing storyline context is cached to M^{SC} :

$$\Delta M^{SC} = g_{s(i-1)} c_{\text{out}}^{\text{story} \top} \quad (5)$$

The item-level context synchronizes to the newly evolving storyline context, as in Eq. 4.

2.4 Storyline return — the key new mechanism

When $s(i) = s_{\text{returning}}$ and storyline $s_{\text{returning}}$ has appeared before, the model reinstates the cached storyline context:

$$c_i^{\text{story}} \leftarrow \lambda \tilde{c}_{s(i)}^{\text{story}} + (1 - \lambda) c_{i-1}^{\text{story}} \quad (6)$$

$$c_i^{\text{item}} \leftarrow c_i^{\text{story}} \quad (7)$$

where $\lambda \in [0, 1]$ is the **reinstatement strength** — the single new parameter introduced by this model. When $\lambda = 1$, returning to a storyline fully restores its context; when $\lambda = 0$, storyline returns behave like ordinary storyline switches.

3. Retrieval dynamics

Retrieval is **identical across all experiments**. The instruction does not enter the retrieval dynamics.

3.1 Context-driven readout

A cue clip i_{cue} is presented. The model retrieves its stored item-level context $c_{i_{\text{cue}}}^{\text{item}}$ from M^{IC} and computes scene activation:

$$a_j = (M^{IC})^\top c_{i_{\text{cue}}}^{\text{item}} \quad (8)$$

Retrieval probabilities are obtained via softmax with inverse temperature k :

$$p(j \mid i_{\text{cue}}) = \frac{\exp(k \cdot a_j)}{\sum_{j'} \exp(k \cdot a_{j'})} \quad (9)$$

This is standard CMR retrieval — the same mechanism Cornell et al. (2024) describe for external cues (the cue's induced context replaces the retrieval context, then search proceeds normally). Applied uniformly across experiments.

3.2 Instruction as a scoring filter

The participant's response is modeled as a sample from $p(j \mid i_{\text{cue}})$. The instruction specifies a **target set** T for each experiment — the set of scenes that count as hits:

- Exp 1 (across-event-within-storyline): $T =$
{scenes in the temporally adjacent event in the original movie}
- Exp 2 (across-storyline): $T =$
{scenes in the temporally adjacent event in the original movie, which belongs to a different story}
- Exp 3 (across-event-bridge): $T =$
{scenes in the next event within the same storyline}

A "hit" occurs when the sampled response falls in T . Responses outside T are generated by the same retrieval process but not scored as correct.

3.3 Empirical support for the scoring-filter account

The scoring-filter account makes a direct, falsifiable prediction about Exp 2 response content. The pilot Exp 2 response annotations confirm this prediction (see §6.2 for details): Exp 2 participants spontaneously produce within-storyline-neighbor responses at rates non-trivial enough to support the account that retrieval produces similar activation distributions regardless of instruction. The Exp 2 $\ll$ Exp 3 gap in *scored* hit rates is therefore a scoring artifact, not a difference in what retrieval produces.

4. Why the model predicts the observed pattern

- **Within-event:** item-level contexts of cue and target are very close; storyline contexts identical. Item-level retrieval (Eq. 8-9) delivers high hit rates.
- **Exp 1 (grouped, across-event-within-storyline):** the event boundary synchronizes item context to storyline context (Eq. 4), separating the target's item context from the cue's drifted item context. But storyline context is temporally continuous between cue and target, which keeps their item-context similarity high. Retrieval recovers the within-storyline neighbor with moderate probability (~ 0.40 hit rate in pilot).
- **Exp 2 (interleaved, across-storyline):** the correct (scored) target is the temporally adjacent event *from a different storyline*. The item-level context at the cue scene (end of a storyline-A chunk) is similar to other storyline-A items, not to the storyline-B target. Retrieval activation on the scored target is low, yielding low hit rates (~ 0.10 - 0.15 in pilot). The same retrieval also produces within-storyline-neighbor activations that aren't scored as hits — these appear in the response content, consistent with the model's prediction.
- **Exp 3 (interleaved, across-event-bridge):** storyline reinstatement at encoding (Eq. 6-7) placed items within a storyline in similar item-context regions, even though separated in

the stream. Retrieval activation on the within-storyline neighbor is high. With the Exp 3 scoring rule (within-storyline neighbor = target), these activations produce hits (~0.40 in pilot, matching Exp 1).

The free parameter λ is identifiable from the Exp1/Exp3 hit-rate ratio. BF01 = 412 in the pilot places λ near the top of its range.

5. Parameter inventory

Parameter	Role	Inherited?	Starting value
β_{enc}	Item-level drift at encoding	Yes (CMR)	0.679
β_{story}	Storyline-level drift	Yes (C&Z β_{list})	0.400
λ	Storyline reinstatement strength (new)	No	free, initialize 0.8
γ_{fc}	Pre- vs. experimental context weight	Yes (CMR)	0.315
k	Retrieval noise / inverse temperature	Yes (CMR)	6.50

Total: 1 new parameter (λ) on top of 4 inherited CMR parameters. Cornell & Zhang's extension introduced 2 parameters (β_{list} , β_{rein}); one (β_{story}) is inherited directly, the other is replaced by λ , which is conceptually related but operates at encoding (on storyline return) rather than at retrieval.

5.1 Parameters from standard CMR that are not used here

Two standard CMR parameters are absent from the inventory because they have no causal role in this paradigm:

- β_{rec} (**retrieval drift rate**). In free recall, β_{rec} governs how the retrieval context drifts toward the encoded context of a just-recalled item, producing within-trial sequential effects such as temporal contiguity across consecutive recalls. In single-response cued recall, each trial consists of one cue and one response: the cue's induced context replaces the retrieval context (the Cornell et al. 2024 mechanism, see §7.1), activation is computed once, one response is sampled, and the trial ends. Nothing happens after the single response that β_{rec} could govern. Including it in the inventory would imply a within-trial dynamic that this paradigm doesn't produce.
- ϵ_d (**stopping-rule rate**). In free recall, participants generate a variable number of responses per trial, and ϵ_d governs the per-attempt stopping probability. In the present paradigm, participants generate exactly one response per cue; there is nothing to stop.

Conditional reintroduction. If the paradigm is extended to include a free-recall component (e.g., the Stage 4 prediction about storyline-based clustering in free recall of the same stimuli, see

§6.4), both parameters become active again: β_{rec} governs within-trial retrieval context drift, and ϵ_d governs recall termination. They would be reintroduced at that point and fit jointly with λ .

6. Simulation plan

6.1 Stage 1: Qualitative proof-of-concept

Use a toy stimulus: 30 scenes, 3 storylines $\times$ 2 events per storyline $\times$ 5 scenes per event, grouped (Exp 1) and interleaved (Exp 2/3) versions. Sweep $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. Check whether any value of λ produces:

- within-event $\gg$ Exp 1 (should emerge from boundary sync at event boundaries)
- Exp 1 $\approx$ Exp 3 under the respective scoring filters
- Exp 2 $\ll$ Exp 1

Diagnostic: identity vs. embedding pre-experimental context (see §1.5). Run Stage 1 twice:

- *Run A* with $M_{\text{pre}}^{FC} = I$ (identity matrix). All scenes orthogonal pre-experimentally. Any Exp 1 $\approx$ Exp 3 equivalence is attributable to the λ reinstatement mechanism alone.
- *Run B* with M_{pre}^{FC} built from within-storyline-similar pseudo-embeddings (toy stimulus) or actual USE embeddings (realistic stimulus). Scenes in the same storyline share pre-experimental content.

Compare the λ values needed to reproduce Exp 1 $\approx$ Exp 3 in each run. Three possible outcomes:

1. *Identical λ needed in both runs*: semantic overlap isn't doing any work; λ carries the effect. The theoretical claim is sharp.
2. *Smaller λ needed in Run B*: semantic overlap contributes some of the equivalence. λ is still needed but less than the identity case suggests. The theoretical claim is partial.
3. *Exp 1 $\approx$ Exp 3 achieved in Run B even at $\lambda = 0$ *: semantic overlap fully accounts for the equivalence. λ is superfluous and the model reduces to "CMR with naturalistic stimuli," with the novelty claim dissolving.

This diagnostic is essential before claiming the λ mechanism is theoretically necessary.

6.2 Stage 2: Validation of the scoring-filter account (completed)

The model's theoretical claim — that retrieval is instruction-blind and instruction acts only as a scoring filter — makes a direct, falsifiable prediction about Exp 2 response content.

Procedure. For each pilot Exp 2 trial, examine the full set of participant responses (not just whether the scored target was hit). Count:

- r_{adj} : rate of responses matching the across-storyline temporally adjacent target (scored as hits in Exp 2)
- r_{bridge} : rate of responses matching the within-storyline neighbor that would be the target in Exp 3 (not scored as hits in Exp 2)

Prediction. If retrieval is instruction-blind, r_{bridge} in Exp 2 should be non-trivial and comparable to the Exp 3 hit rate for structurally matched cues.

Result (pilot). Exp 2 participants produce a decent number of within-storyline-neighbor responses. The rate is substantial enough to support the scoring-filter account: retrieval is generating bridge-style responses in Exp 2 just as it does in Exp 3, but the Exp 2 scoring rule doesn't count them as hits. The Exp 2 $\ll$ Exp 3 gap in *scored* hit rates is therefore a scoring artifact, not a difference in what retrieval produces.

This validates the pure-encoding formulation and removes the motivation for an instruction-dependent retrieval mechanism. Any model with an instruction-modulated retrieval route would have had to predict r_{bridge} (Exp 2) $\ll$ Exp 3 hit rate, which is contradicted by the data.

6.3 Stage 3: Fit to pilot data

Fix the CMR-inherited parameters at Cornell & Zhang's published values (see §5). Fit λ alone to hit rates and lag distributions across the four conditions. With 1 free parameter against 4+ target statistics (hit rates per condition, plus lag-distribution shape), the model is strongly constrained — a successful fit is informative rather than a curve-fitting artifact.

For the full preregistered sample, also fit to content-similarity measures (the planned pairwise cosine-similarity analysis) in addition to hit rates, to further constrain λ .

6.4 Stage 4: Out-of-sample predictions

The model should produce without additional parameters:

- **Temporal symmetry** in Exp 1 and Exp 3 but not Exp 2 (already observed in pilot; non-trivial match).
- **Lag distributions:** Exp 1 and Exp 3 comparable; Exp 2 flatter.
- **Content similarity:** Exp 1 and Exp 3 participants draw from the same region of item-context space, yielding comparable pairwise cosine similarities within- and between-experiments.
- **Free-recall clustering** (if collected): storyline-based clustering should emerge under interleaved encoding even without a within-storyline instruction, because storyline context organizes the encoded representation regardless of retrieval framing.

7. Implementation notes

Cornell & Zhang's code is a natural starting point. Rename list-level context as storyline-level context, treat within-storyline event boundaries with the existing boundary-sync rule, and add Eq. 6-7 at storyline returns. Retrieval dynamics are unchanged from standard CMR.

For naturalistic stimuli, use USE embeddings of subtitle text as the pre-experimental input c_i^{IN} , normalized to unit norm. For fast debugging, use one-hot inputs with a toy stimulus structure.

7.1 Relation to Cornell et al. (2024) cueing mechanism

The retrieval mechanism is structurally identical to the external-cue mechanism in Cornell et al. (2024, *Psychological Science*). Their model handles mid-retrieval external cues as follows: when the participant is stuck, an external cue is presented, and the current context is replaced by the cue's induced context ($c_t = c_{\text{cue}}$, equivalent to setting $\beta = 1$ with the cue as input). After this replacement, retrieval proceeds normally with the same parameters and process as pre-cue retrieval.

The present model applies the same principle at the *start* of cued recall rather than mid-retrieval: the cue clip's induced context becomes the retrieval context, and standard CMR search produces an activation distribution over scenes. Instructions do not enter the retrieval dynamics — they only filter which responses count as hits.

This framing concentrates the model's novelty on one encoding-side mechanism (λ , storyline reinstatement) and inherits the retrieval mechanism from established theory.

8. Key differences from Cornell & Zhang

Feature	Cornell & Zhang	This model
Context levels	2 (item, list)	2 (item, storyline)
Boundary mechanisms	Full sync at list boundary	Full sync at event & storyline boundaries (inherited)
Reinstatement	At retrieval, from M^{lists}	At encoding , on storyline return (new)
Retrieval route	Item-level	Item-level (inherited)
Domain	List learning, final free recall	Naturalistic narrative cued recall
New parameters vs. CMR	$2 (\beta_{\text{list}}, \beta_{\text{rein}})$	$1 (\lambda)$

The substantive novelty is encoding-time reinstatement on storyline resumption, which Cornell & Zhang don't face because their lists don't interleave. In naturalistic film, storylines *do* interleave, and the model has to account for how viewers maintain storyline-specific representations across interruptions. Every other mechanism — drift dynamics, boundary synchronization, context-driven retrieval — is inherited unchanged.

9. Risks and open questions

- **Pre-experimental semantic overlap may absorb λ 's explanatory role** (see §1.5 and §6.1 diagnostic). If USE embeddings of within-storyline scenes are already highly similar, the apparent storyline-reinstatement effect could be explained by pre-experimental content

overlap alone, with $\lambda = 0$. The Stage 1 identity-vs-embedding comparison is designed to detect this and is essential before claiming the λ mechanism is theoretically necessary.

- **λ identifiability:** one free parameter against four hit rates plus lag distributions plus content-similarity measures should be well-identified, but the full preregistered sample size is what actually resolves this. Stage 3 fit with pilot data gives a preliminary estimate; the full sample gives confidence intervals.
 - **$\lambda = 1$ plausibility:** λ must be close to 1 to make Exp 1 $\approx$ Exp 3 exactly. If the true effect is slightly Exp 1 $>$ Exp 3 (not ruled out with $n = 17$ per experiment despite $BF_{01} = 412$), $\lambda < 1$ and the model is still viable with a slight residual difference. The full preregistered sample will adjudicate.
 - **Storyline detection:** the model assumes storyline labels are known. Viewers have to infer them. A future extension could use Bayesian surprise over a language model to detect storyline boundaries online (as Cornell & Zhang suggest in their Discussion).
 - **Optional extension — partial event-boundary reset.** If stage-1 simulations show the full-sync event boundary produces too small a within-event vs across-event-within-storyline gap, the event boundary sync can be relaxed to a partial reset with an α parameter. This should be introduced only if needed.
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Next steps

1. Python code for the hierarchical extension (building on the existing `cmr.py`).
2. Stage 1 simulation with identity-vs-embedding diagnostic.
3. Stage 3 fit of λ to pilot hit rates.
4. Methods / Discussion text.